Ng Ri Chi

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Work Experience

AI Engineer at AI Singapore November 2025 to present

Question setting for a national level coding and AI competition in line with paedagogical best practices

- Researched and developed guidelines on paedagogical best practices when setting questions
- Creation of agentic pipeline using PydanticAI for generation of academic focused questions and answers.
- Editing and shortlisting of generated MCQ
- Creation and testing of coding questions to introduce candidates to huggingface, pytorch and related libraries
- Mentorship of student helper to aid in question setting and testing

Development of competition infrastructure for AI coding competitions

- Led a team to develop post-competition analysis pipeline which runs Jupyter Notebooks through OtterGrader for software tests and an agent built with Pydantic AI for subjective feedback. This was used for the National Olympiad in AI 2026 finals, where I also edited notebooks to make them compatible with the pipeline. 150 scripts across 3 questions were automatically analysed using this pipeline.
- Optimised use of GPUs through the use of Jupyter Enterprise Gateway to allow use of more powerful GPU instance in other countries to run GPU intensive code while maintaining low latency instances locally for JupyterLab interaction running on Docker

Research Assistant at AI Singapore March 2025 to November 2025

AI development and deployment of a picture description tutor for the Education Grand Challenge in partnership with the Ministry of Education

- Benchmarked local multiple multimodal LLMs using HuggingFace Transformers library in AWS instances and closed LLMs using OpenAI API on tasks like scene detection and future action prediction
- Worked closely with product owner to navigate demanding requirements to develop an agentic framework using Pydantic AI with tool calling for the Visual Dialogue Engine
- Integration of agentic model into backend powered by FastAPI

Research Assistant at SUTD Social AI Studio August 2023 to February 2025

First author and project lead of two papers relating to responsible AI evaluation, where I created a speech benchmark dataset for Singapore (WOAH 2024) and Southeast Asia (TALLIP 2026). In total close to 80,000 test cases were created.

- Used AWS Elastic Beanstalk and RDS for PostgreSQL to host a Flask based translation and annotation web portal
- Supervised human editing of closed source LLM translated templates by native speakers and subsequent human annotation of test cases by language and culture experts. Used advanced techniques like inter-annotator agreement to ensure data reliability.
- Used HuggingFace's Transformer library to fine-tuned and inferenced local LLMs with advanced engineered prompts for generation and benchmarking of completed dataset

Research Assistant at SUTD Aviation Studies Institute September 2022 to March 2023

Identified factors that increased fuel burn rate using machine learning and advanced data analysis

- Engineered time series data with NumPy and pandas in Python
- Trained XGBoost model using scikit-learn and tuned hyperparameters used
- Distilled and visualised results using SHAP and identified significant contributory factors

Honours Project

Batchsmarter with Kuehne+Nagel SUTD Capstone

Developed a mixed-integer optimisation model that streamline order picking process by grouping similar orders together, while ensuring constrains like packing volume, weight and urgency. The model resulted in estimated time savings of up to 10%.

Education

2019 - 2023 **Singapore University of Technology and Design (SUTD)** *Honours with Distinction, GPA: 4.43/5.00*
Bachelor of Engineering (Engineering Systems Design) with Honours
Singapore Management University (SMU) *Magma Cum Laude, GPA: 3.60/4.00*
Bachelor of Business Management
SUTD-SMU Dual Degree Programme Scholarship Tier 1